

# Phase 1C Completion Summary: Stripe Billing & Super Admin Settings

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**Date:** August 21, 2026

**Version:** 1.0.0

**Status:**  COMPLETE

## Overview

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Phase 1C added the monetization layer to BaseCenter.ai: a secure Super Admin (SA) settings area for managing Stripe API keys, full Stripe billing integration (customers, checkout, subscriptions, webhooks), module catalog seeding, and a frontend admin console for configuring everything without touching code.

This directly implements the BaseCenter business model — one free module forever, additional modules at **\$5/month each (10 seats per module)**, plus a free Website Builder.

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## Deliverables Completed

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### 1. Super Admin Settings Area

The centerpiece of this phase: Super Admins can input and manage Stripe API keys through a secure UI — no `.env` edits or redeployments required.

- **Backend:** `GET/PUT /api/v1/admin/settings/stripe` , `POST /api/v1/admin/settings/stripe/test`
- **Frontend:** `/admin` (SA login) → `/admin/settings` (Stripe Settings console)
- Fields managed: **Mode** (test/live), **Publishable Key**, **Secret Key**, **Webhook Signing Secret**
- Live connection status indicator: Not configured → Configured (test mode) / (live mode)
- **Test Connection** button validates the saved secret key against the Stripe API

### 2. Encrypted Secret Storage

**Location:** `backend/app/core/encryption.py` , `backend/app/models/setting.py`

- Secret values (secret key, webhook secret) are **encrypted at rest** using Fernet (symmetric AES) before hitting the database.
- Encryption key derived via SHA-256 from `SETTINGS_ENCRYPTION_KEY` .
- On read, secrets are **masked** — only the last 4 characters are shown (e.g. `.....y456` ).
- Blank fields on update **preserve** the existing stored value, so keys never need re-entry.
- Non-secret values (publishable key, mode) stored as plaintext.

 **Production note:** `SETTINGS_ENCRYPTION_KEY` must be set to a stable secret in production. Changing it after keys are saved makes previously stored secrets undecryptable.

### 3. Stripe Service Layer

**Location:** `backend/app/services/stripe_service.py`

- Reads the secret key **from the database at call time** (not from env), so SA changes take effect immediately.
- `test_connection` — validates credentials
- `ensure_stripe_product_and_price` — creates/links Stripe Product + Price per module
- `get_or_create_customer` — maps tenants to Stripe customers
- `create_checkout_session` — hosted checkout for a module subscription
- `cancel_subscription` — cancels an active subscription
- `construct_webhook_event` — verifies webhook signatures
- Raises `StripeNotConfiguredError` when keys are absent — endpoints fail gracefully.

### 4. Module Catalog & Seeding

**Location:** `backend/app/db/seed.py` , `backend/app/models/module.py`

Seeded **11 modules** with pricing metadata ( `is_free_eligible` , `stripe_product_id` , `stripe_price_id` ):

| #  | Module               | Slug               | Price  | Free-eligible                                                                         |
|----|----------------------|--------------------|--------|---------------------------------------------------------------------------------------|
| 1  | Invoice & Milestones | invoice-milestones | \$5/mo |   |
| 2  | Contracting          | contracting        | \$5/mo |  |
| 3  | Help Ticket          | help-ticket        | \$5/mo |  |
| 4  | Knowledge Base       | knowledge-base     | \$5/mo |  |
| 5  | Data Collection      | data-collection    | \$5/mo |  |
| 6  | Project Management   | project-management | \$5/mo |  |
| 7  | File Management      | file-management    | \$5/mo |  |
| 8  | Training LMS         | training-lms       | \$5/mo |  |
| 9  | Accounting           | accounting         | \$5/mo |  |
| 10 | CRM Plus             | crm-plus           | \$5/mo |  |
| 11 | Website Builder      | website-builder    | Free   | —                                                                                     |

The **Sync Modules to Stripe** action (SA console + `POST /admin/modules/sync-stripe` ) creates the corresponding Stripe Products/Prices for all paid modules in one click.

## 5. Billing API Endpoints

**Location:** `backend/app/api/v1/`

- `GET /api/v1/modules` — public list of active modules
- `GET /api/v1/admin/modules` — SA module list with sync status
- `POST /api/v1/admin/modules/sync-stripe` — create Stripe products/prices
- `GET /api/v1/subscriptions` — list a tenant's subscriptions
- `POST /api/v1/subscriptions/checkout` — start a Stripe Checkout session
- `POST /api/v1/subscriptions/{id}/cancel` — cancel a subscription

## 6. Stripe Webhook Handler

**Location:** `backend/app/api/v1/webhooks.py`

- `POST /api/v1/webhooks/stripe` — signature-verified endpoint
- Handles: `checkout.session.completed`, `customer.subscription.created`, `customer.subscription.updated`, `customer.subscription.deleted`
- Keeps local subscription state in sync with Stripe.

## 7. Role-Based Access Control

**Location:** `backend/app/api/deps.py`

- `get_current_superuser` dependency guards all `/admin/*` routes.
- Non-super-admin users receive **HTTP 403**.

## 8. Frontend Admin Console

**Location:** `app/app/admin/`, `app/components/admin/`, `app/lib/api.ts`

- `app/admin/page.tsx` — SA login (auto-redirects if already authenticated)
- `components/admin/AdminShell.tsx` — sidebar layout, auth guard, logout
- `app/admin/settings/page.tsx` — Stripe Settings console (status, keys form, module sync)
- `lib/api.ts` — typed API client, JWT stored in `localStorage`

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## Database Changes

Alembic migration `ca8f070d09de` added:

- `platform_settings` table — key/value store with `is_secret` flag (encrypted values)
- `modules` columns — `is_free_eligible`, `stripe_product_id`, `stripe_price_id`

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## New Dependencies

| Package                   | Version | Purpose                       |
|---------------------------|---------|-------------------------------|
| <code>stripe</code>       | 11.3.0  | Stripe API SDK                |
| <code>cryptography</code> | 43.0.3  | Fernet encryption for secrets |

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## Verification & Testing

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All items below were tested successfully:

- ☒ Frontend production build ( `next build` ) — compiles, typechecks, 5 routes generated
  - ☒ Public modules endpoint returns all 11 modules
  - ☒ SA login issues JWT; `/admin` redirects to settings when authenticated
  - ☒ Save Stripe keys → status flips to Configured (test mode)
  - ☒ Secrets **encrypted at rest** (Fernet tokens in DB); publishable key + mode plaintext
  - ☒ Secrets **masked** on read (last-4 only)
  - ☒ Blank re-submit **preserves** existing secret keys
  - ☒ Test Connection with an invalid key fails gracefully (“Invalid Stripe secret key”)
  - ☒ Non-admin user blocked with **HTTP 403**
  - ☒ Module sync UI lists 10 paid modules (Not synced) + Website Builder (Free)
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## How to Use (Super Admin)

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1. Start the backend and frontend.
  2. Go to `/admin` and sign in with the super admin account.
  3. On **Stripe Settings**, choose **Mode**, paste your **Publishable Key**, **Secret Key**, and **Webhook Signing Secret**, then **Save Settings**.
  4. Click **Test Connection** to confirm the credentials work.
  5. Click **Sync Modules to Stripe** to create Products/Prices for all paid modules.
  6. In the Stripe Dashboard, point your webhook endpoint at `/api/v1/webhooks/stripe` and paste the signing secret back into the settings.
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## What's Next

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- **Phase 1D** — Full Super Admin portal: tenant management, subscription/seat administration, usage analytics.
  - **Customer-facing billing** — module marketplace, “add module” checkout flow, billing history in the tenant dashboard.
  - **Phase 2+** — Build out the 10 business modules and the AI setup wizard.
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Generated for BaseCenter.ai — Phase 1C.